

RAG Document Chatbot

Fully local retrieval-augmented document Q&A;

Portfolio

Masters

Repo / Path	asadbek/AI ML/3-project
Domain	LLM / retrieval-augmented generation
Stack	LangChain + ChromaDB, Ollama (nomic-embed-text + llama3.2), FastAPI; React 19 + TS + Tailwind v4
Status	Working - ~85% complete
Live / Deploy	Not publicly deployed

Overview

A fully local RAG chatbot: upload PDFs / text, ask questions, and get cited answers - all running locally with no external API.

Key Features & Technical Highlights

- Complete RAG pipeline: chunk -> embed -> persist -> similarity search -> grounded prompt -> generate.
- Persistent Chroma vector store with live add/delete of documents.
- Local embeddings + LLM via Ollama; typed REST contract.

Engineering Evidence & Metrics

- Real end-to-end RAG system with a pre-built vector store and sample docs.
- Engineering-heavy (pipeline design) using pretrained models.

Why It Matters (application / interview talking points)

- Demonstrates practical LLM engineering and retrieval design - relevant to applied-AI programs.